

21. Let $S = \{\sin^2 2\theta : (\sin^4 \theta + \cos^4 \theta)x^2 + (\sin 2\theta)x + (\sin^6 \theta + \cos^6 \theta) = 0 \text{ has real roots}\}$. If α and β be the smallest and largest elements of the set S , respectively, then $3((\alpha - 2)^2 + (\beta - 1)^2)$ equals _____.

माना $S = \{\sin^2 2\theta : (\sin^4 \theta + \cos^4 \theta)x^2 + (\sin 2\theta)x + (\sin^6 \theta + \cos^6 \theta) = 0 \text{ के वास्तविक मूल हैं}\}$ हैं। यदि α तथा β समुच्चय S के क्रमशः सबसे छोटा तथा सबसे बड़ा अवयव हैं, तो $3((\alpha - 2)^2 + (\beta - 1)^2)$ बराबर है _____

Ans. 4

Sol. $D \geq 0$

$$\Rightarrow \sin^2 2\theta - 4(\sin^4 \theta + \cos^4 \theta)(\sin^6 \theta + \cos^6 \theta) \geq 0$$

$$\Rightarrow t - 4\left(1 - \frac{t}{2}\right)\left(1 - \frac{3t}{4}\right) \geq 0 \quad (\text{Let } \sin^2 2\theta = t)$$

$$\Rightarrow 3t^2 - 12t + 8 \leq 0$$

$$\Rightarrow t \in \left[2 - \frac{2}{\sqrt{3}}, 2 + \frac{2}{\sqrt{3}}\right] \quad \text{_____ (1)}$$

$$\text{But } t = \sin^2 2\theta \in [0, 1] \quad \text{_____ (2)}$$

$$(A) \cap (2) t \in \left[2 - \frac{2}{\sqrt{3}}, 1\right]$$

$$\alpha = 2 - \frac{2}{\sqrt{3}}, \beta = 1$$

$$3((\alpha - 2)^2 + (\beta - 1)^2) = 4$$

22. The number of solutions of $\sin^2 x + (2 + 2x - x^2) \sin x - 3(x - 1)^2 = 0$, where $-\pi \leq x \leq \pi$, is _____.

$\sin^2 x + (2 + 2x - x^2) \sin x - 3(x - 1)^2 = 0$, जहाँ $-\pi \leq x \leq \pi$ के हलों की संख्या है _____

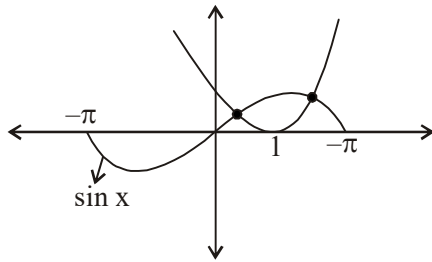
Ans. 2

Sol. $\sin^2 x + (3 - (x - 1)^2) \sin x - 3(x - 1)^2 = 0$

$$\sin^2 x + 3 \sin x - (x - 1)^2 \sin x - 3(x - 1)^2 = 0$$

$$\sin x (\sin x + 3) - (x - 1)^2 [\sin x + 3] = 0$$

$$\underbrace{(\sin x + 3)}_{\substack{\sin x \neq -3 \\ \downarrow \\ \text{not possible}}} \underbrace{(\sin x - (x - 1)^2)}_{\sin x = (x - 1)^2} = 0$$



There are two intersections between this graph.

So, number of solution will be 2.

23. Let the set of all $a \in \mathbb{R}$ such that the equation $\cos 2x + a \sin x = 2a - 7$ has a solution be $[p, q]$ and

$$r = \tan 9^\circ - \tan 27^\circ - \frac{1}{\cot 63^\circ} + \tan 81^\circ, \text{ then } pqr \text{ is equal to } \underline{\hspace{2cm}}.$$

माना $a \in \mathbb{R}$ के सभी मानों, जिनके लिए समीकरण $\cos 2x + a \sin x = 2a - 7$ का एक हल है, का समुच्चय $[p, q]$ है तथा

$$r = \tan 9^\circ - \tan 27^\circ - \frac{1}{\cot 63^\circ} + \tan 81^\circ \text{ है। तो } pqr \text{ बराबर है } \underline{\hspace{2cm}}।$$

Ans. 48

Sol. $\cos 2x + a \sin x = 2a - 7$
 $a(\sin x - 2) = 2(\sin x - 2)(\sin x + 2)$
 $\sin x = 2, a = 2(\sin x + 2)$
 $\Rightarrow a \in [2, 6]$
 $p = 2 \quad q = 6$
 $r = \tan 9^\circ + \cot 9^\circ - \tan 27^\circ - \cot 27^\circ$
 $r = \frac{1}{\sin 9^\circ \cdot \cos 9^\circ} - \frac{1}{\sin 27^\circ \cdot \cos 27^\circ}$
 $= 2 \left[\frac{4}{\sqrt{5}-1} - \frac{4}{\sqrt{5}+1} \right]$
 $r = 4$
 $p \cdot q \cdot r = 2 \times 6 \times 4 = 48$